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Unit 4 - BUILDING THE PROJECT

PDF Documents

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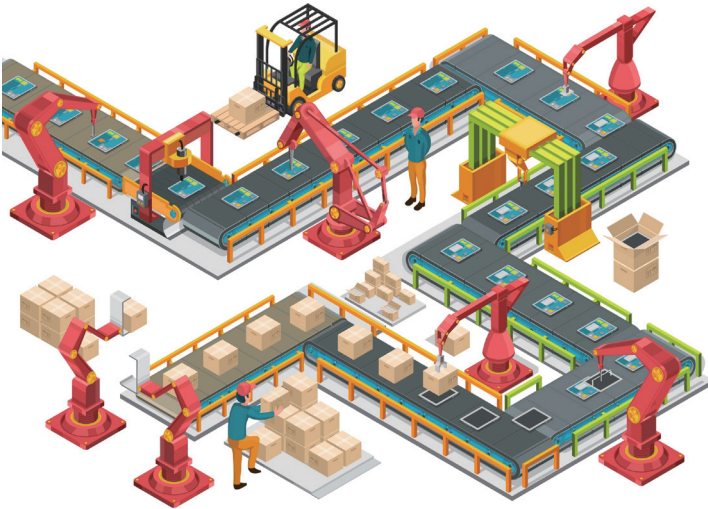


UNIT 4 BUILDING THE PROJECT

Unit Introduction

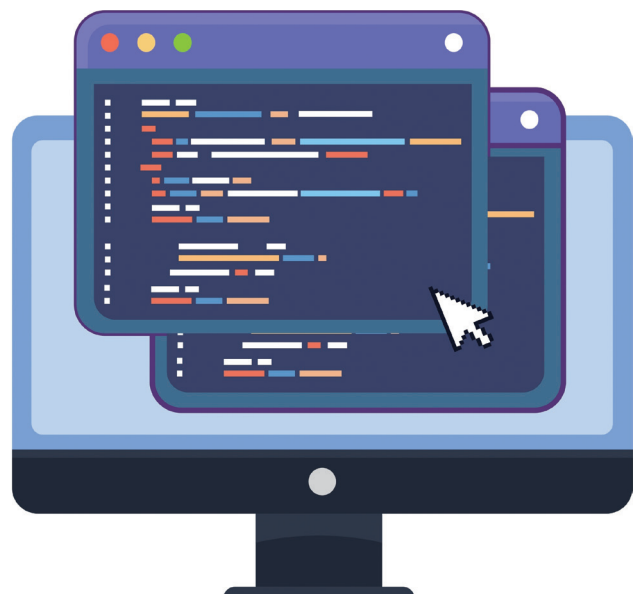
In Unit 4, you will begin the hands-on process of building and coding your smart device. This is where all your planning and preparation come to life. You will use the MODI modules, your detailed algorithms, and flowcharts to assemble and program your smart device. This unit will guide you through the steps of constructing your project, ensuring that each component works together seamlessly.

STEP 1) ASSEMBLING THE HARDWARE



Connecting the hardware required and chosen for your creation is crucial. Without proper connections, your device will not work as intended. Therefore, connect the hardware, considering your desired physical architecture and its functions!

STEP 2) PROGRAMMING THE DEVICE



Programming is another important part!

Even if your physical architecture is well-built, incorrect programming can cause malfunctions. Your logical thinking skills are needed!

Assembling the Hardware

Steps Assembling the Hardware

a) Gather Materials:

- Collect all the MODI modules you need for your project, such as sensors, actuators, and the battery module.
- Gather any additional materials like wires, connectors, and mounting tools.

b) Connect the Modules:

- Follow your flowchart and algorithm to connect the MODI modules in the correct order.
- Ensure all connections are secure and that the modules are positioned correctly.

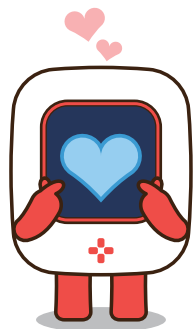
c) Check Connections:

- Double-check all connections to make sure there are no loose or incorrect connections.
- Test each connection to ensure it is working properly.

Example



Hardware connection is important. To turn your ideas into reality, think about how the hardware should be assembled.





2) Programming the Device

a) Write the Code:

- Use your algorithms as a guide to write the code for your smart device.
- Program each module according to the detailed steps in your algorithm.

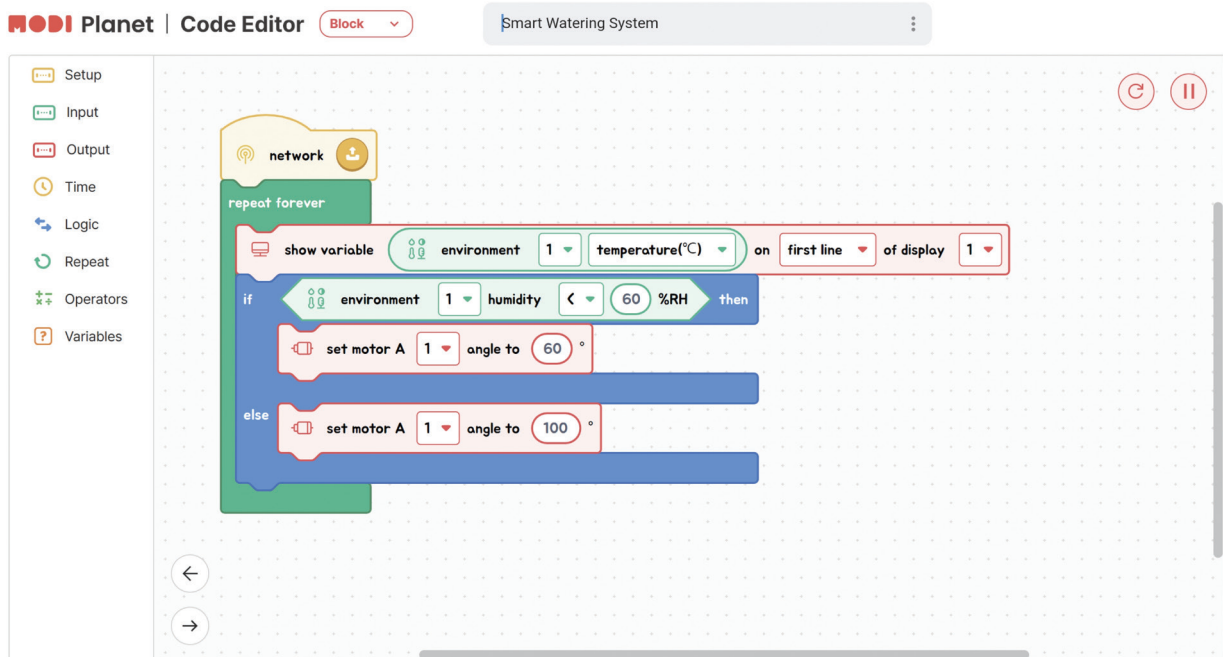
b) Upload the Code:

- Use MODI Studio to upload your code to the MODI modules.
- Ensure that each module receives the correct code and that the overall program is functioning as intended.

c) Test the Code:

- Run the program to see if the device performs as expected.
- Observe the behavior of each module and make sure they are working together correctly.

Example



Do you remember how we wrote codes based on different flowcharts in the previous chapters? I know it is challenging, but remember that you have done well many times! Be confident! If it is not working, fix it!